Young Hyun Kim

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RESEARCH INTERESTS

Human cardiac tissue engineering and organoid models, chamber-specific electromechanical function, mechanobiology, and quantitative phenotyping of hiPSC-derived cardiac systems.

EDUCATION

Sejong University Seoul, Republic of Korea
M.S. in Integrative Bioscience and Biotechnology **Aug 2026**
GPA: 4.07/4.5
Thesis: *Generation of Chamber-Specific 3D Cardiac Organoids and an Integrated Electro-Mechanical Optical Analysis Platform*.

B.Eng. in Integrative Bioscience and Biotechnology **Aug 2023**
GPA: 3.50/4.5

Université Catholique de Lille Lille, France
Exchange Student, ISA Lille (Master's-level coursework) Sep 2019 – Dec 2019

PUBLICATIONS

1. **Kim, Y. H.**, Son, Y. H., Choi, Y., Kim, M. S., Park, S. J., & Lee, K. Y. (2026). Advanced in vitro cardiac models for drug evaluation: Integration of organoids, engineered tissues, and microphysiological systems. *Microsystems & Nanoengineering*, 12(1), 162. **(Review)**
2. Kim, M. S., **Kim, Y. H.**, Choi, Y., & Lee, K. Y. (2025). Biohybrid actuators in robotics: recent trends and future perspectives of skeletal and cardiac muscle integration. *npj Robotics*, 3(1), 37. **(Review)**
3. Kim, Y. S., Lee, M., Cheon, Y. J., Choi, H. S., Choi, B. J., **Kim, Y. H.**, Lee, K. Y., & Ryu, C. J. (2026). NPB35, a monoclonal antibody against a glycan-dependent conformational epitope on CD147, collapses the CD147–CD98hc–MCT1/MCT4 module and suppresses hepatocellular carcinoma growth. *Cancer Immunology, Immunotherapy*. **(*Revision requested*)**

RESEARCH EXPERIENCE

Biological Systems Engineering Laboratory, Sejong University Seoul, Korea
Dept. of Integrative Bioscience and Biotechnology · Prof. Keel Yong Lee
Research Assistant Sep 2026 – Present

Integrated Electromechanical Stimulation and Incubation System

- Designing a new system to coordinate electrical pacing and mechanical stretch in an integrated incubation chamber; defining culture and stimulation requirements and coordinating fabrication for cardiac tissue studies.

Dual-Camera Fluorescence/Brightfield Imaging System

- Extending my master's work with a dichroic beam splitter for simultaneous Fluo-4 calcium imaging and brightfield motion recording in cardiac organoids. Planning paired optical and microelectrode array (MEA) recordings to relate calcium dynamics and contraction to electrical activity in the same tissue.

Graduate Research Assistant Sep 2023 – Aug 2026

Thesis Project: Chamber-Specific Cardiac Organoids and Integrated Functional Analysis

- Generated atrial- and ventricular-like 3D cardiac organoids using a biphasic 2D-to-3D differentiation workflow with stage-specific Wnt and retinoic-acid modulation.
- Assessed lineage specification by qRT-PCR and cardiomyocyte purity by cTnT flow cytometry in monolayer cultures; characterized 3D organoid structure and marker expression by immunofluorescence.
- Developed iCAMAnalyzer, a Python-based platform integrating calcium-transient analysis, calcium-based activation mapping, and label-free motion analysis, and applied it to compare atrial- and ventricular-like cardiac organoids.
- Benchmarked activation mapping against synthetic ground-truth videos (CV recovery $R^2 > 0.99$ for tested planar and radial waves) and assessed sensitivity to frame rate, noise, and spatial resolution.
- Identified faster, shorter calcium transients and higher spontaneous beating rates in atrial-like versus ventricular-like organoids; calcium-derived propagation velocities were comparable.
- Developed a MATLAB morphology analyzer and compared diameter measurements with manual ImageJ analysis ($R^2 = 0.973$; $n = 27$).
- Demonstrated parallel optical analysis on a nine-organoid micro-pillar array and tested motion analysis on organoid samples provided by Konkuk University researchers.

Additional Research & Collaborations

- Investigated spinner-flask culture conditions for scaling hiPSC-derived cardiomyocyte production and organoid fabrication.
- Differentiated and supplied hiPSC-derived cardiomyocytes and jointly performed flow cytometry characterization for a collaborative study of CD147 targeting in hepatocellular carcinoma.

Undergraduate Research Assistant Oct 2022 – Aug 2023

- Established the laboratory's initial hiPSC-derived cardiac organoid fabrication protocol, creating the foundation for subsequent organoid research.
- Fabricated C2C12-based 3D skeletal muscle tissues and applied cyclic mechanical stretch to assess tissue maturation.
- Joined as one of the laboratory's first three members and helped establish equipment, core experimental protocols, and safety procedures.

LABORATORY TOURS

Harvard University and Emory University, USA **Sep 2024**

- Visited the laboratories of Prof. Kit Parker (Harvard) and Prof. Sung Jin Park (Emory).

CONFERENCE PRESENTATIONS

- Poster; Korea BioChip Society (Nov 2025, Jeju) — Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis
- Poster; 2025 Summer Symposium on Biological Convergence Engineering (Sep 2025, Sejong University) — Chamber-Specific Cardiac Organoid Platform with Micropillar-Assisted Calcium and Optical Signal Analysis

TEACHING AND MENTORING

Undergraduate Research Mentor, SURF Program Jul 2026 – Aug 2026

- Mentored 4 undergraduate researchers in cell culture, experimental planning, confocal imaging, and final project presentations.

Teaching Assistant, Biochemistry Laboratory Jul 2023 – Dec 2024

- Guided 120 students through antibody-production workflows, from plasmid transformation to protein purification and Western blotting; prepared teaching materials.

*Teaching Assistant***,** Microbiology Laboratory Jan 2024 – Jun 2024

- Instructed 40 students in bacterial culture, Gram staining, biochemical assays, and API-based identification.

*Peer Research Mentor***,** DoLearn-DoRun Program Sep 2023 – Dec 2023

- Mentored 2 undergraduates investigating myotube alignment on patterned substrates; authored the proposal and directed experiments.

SKILLS

Cell & Tissue Engineering: hiPSC culture; chamber-specific cardiomyocyte differentiation; 3D cardiac organoid and skeletal muscle tissue engineering.

Molecular & Cellular Assays: qRT-PCR; flow cytometry; immunofluorescence staining.

Imaging & Functional Analysis: confocal microscopy; Fluo-4–based calcium-transient imaging; calcium-based activation mapping; brightfield-based contractility analysis; microelectrode array (MEA) recording.

Design & Fabrication: Rhino; FDM and SLA 3D printing; laser cutting.

Programming & Computational Analysis: Python; MATLAB; COMSOL Multiphysics (hyperelasticity, FSI, cardiac electrophysiology); ImageJ; GraphPad Prism.

Scientific Communication: Adobe Illustrator; EndNote.

REFERENCES

Keel Yong Lee, PhD

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Relationship: Master's thesis advisor and current research supervisor